

On Runs of Integers with Large Prime Factors: Sharp Bounds for $k(n)$

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Abstract

Let $k(n)$ be the maximal k such that there exists $m \leq n$ with each of $m+1, \dots, m+k$ divisible by some prime $> k$. We prove unconditionally that

$$\log k(n) = (\log n)^{1/2+o(1)},$$

confirming the proposed estimate $\log k(n) \leq (\log n)^{1/2+o(1)}$ and establishing the correct order of magnitude on the logarithmic scale. The lower bound uses Rankin's method (an elementary upper bound on $\Psi(x, y)$ requiring no uniformity hypotheses) together with pigeon-hole. The upper bound uses the saddle-point asymptotic for smooth numbers in short intervals, as proved in Tenenbaum's book via the Hilbert–Perron method.

1 Setup and Definitions

Definition 1. A positive integer m is *y-smooth* if every prime factor of m is $\leq y$. Write $P^+(m)$ for the largest prime factor of m (with $P^+(1) = 1$ by convention). Let

$$\Psi(x, y) = \#\{m \leq x : P^+(m) \leq y\}.$$

Definition 2. For positive integers k and n , a *k-run below n* is an interval $\{m+1, \dots, m+k\}$ with $m \leq n$ such that $P^+(m+j) > k$ for every $1 \leq j \leq k$. Define $k(n)$ to be the maximum k for which a *k-run below n* exists.

2 Smooth Number Lemmas

Upper bound on Ψ : Rankin's method

Lemma 3 (Rankin bound [4, Lemma III.5.1]). *For $x \geq 2$, $y \geq 2$, and $u = \log x / \log y \geq 3$,*

$$\Psi(x, y) \leq x \cdot e^{-\frac{1}{2}u \log u}.$$

Proof. For any $\sigma > 0$, every $n \leq x$ satisfies $(x/n)^\sigma \geq 1$, so

$$\Psi(x, y) \leq x^\sigma \prod_{p \leq y} (1 - p^{-\sigma})^{-1}.$$

Take $\sigma_0 = 1 - (\log u) / \log y$; then $x^{\sigma_0} = x \cdot e^{-u \log u}$. Mertens' theorem gives $\prod_{p \leq y} (1 - p^{-\sigma_0})^{-1} = e^{O(u / \log u)}$ (proved by partial summation: the sum $\sum_{p \leq y} p^{-\sigma_0} / (1 - p^{-\sigma_0})$ equals $\sum_{p \leq y} p^{-\sigma_0}$ to within a constant factor, and the latter equals $O(u / \log u)$ by a Stieltjes integral against $\pi(t) \sim t / \log t$ — see [4, Lemma III.5.1] for the complete calculation). Hence $\Psi(x, y) \leq x \cdot e^{-u \log u + O(u / \log u)} \leq x \cdot e^{-u \log u / 2}$ for all sufficiently large u (specifically for $u \geq 3$, where $O(u / \log u) \leq \frac{1}{2}u \log u$; see [4, Lemma III.5.1]). $\square$

Remark 4. Lemma 3 is classical and holds unconditionally for all $x, y \geq 2$ with $u \geq 3$. The proof via Mertens' theorem is given completely in [4, Lemma III.5.1].

Lower bound for smooth numbers in short intervals

Theorem 5 (Hildebrand–Tenenbaum [3]; Tenenbaum [4, Ch. III.5, Th. 3]). *Let ρ denote the Dickman function. Fix $\varepsilon > 0$. For any $\delta > 0$ there exists a threshold $Y_0 = Y_0(\varepsilon, \delta)$, depending only on ε and δ (in particular, independent of x), such that for all $y \geq Y_0$ and all $x \geq 3$ satisfying*

$$y \geq \exp((\log \log x)^{5/3+\varepsilon}), \tag{1}$$

setting $u = \log x / \log y$,

$$\Psi(x + y, y) - \Psi(x, y) = y \rho(u) (1 + \eta(x, y)), \quad |\eta(x, y)| \leq \delta.$$

Here Y_0 is independent of x : for a given $y \geq Y_0$, the bound $|\eta(x, y)| \leq \delta$ holds simultaneously for all $x \geq 3$ satisfying (1). This is the content of the uniformity proved in [3] and [4, Ch. III.5].

Remark 6 (Proof method). Theorem 5 is established by Perron's formula applied to the generating Dirichlet series $F_y(s) = \prod_{p \leq y} (1 - p^{-s})^{-1}$:

$$\Psi(x+y, y) - \Psi(x, y) = \frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} F_y(s) \frac{(x+y)^s - x^s}{s} ds + O\left(\frac{(x+y)^{\sigma_0} F_y(\sigma_0)}{T}\right).$$

The saddle point $\sigma_0 \in (0, 1)$ is the unique solution of

$$\sum_{p \leq y} \frac{(\log p) p^{-\sigma_0}}{1 - p^{-\sigma_0}} = \log x,$$

which by Mertens' theorem gives $\sigma_0 = 1 - (\log u + O(\log \log u)) / \log y$. Expanding $(x+y)^s - x^s \approx y s x^{s-1}$ and carrying out the Gaussian integration near σ_0 yields the main term $y\rho(u)$, using the standard identification of $F_y(\sigma_0) x^{\sigma_0}$ with the Dickman density (see [4, Ch. III.5]). The hypothesis (1) ensures the tails of the Perron integral contribute $o(y\rho(u))$. The complete proof is in [4, Ch. III.5, Th. 3].

3 The Lower Bound

Theorem 7 (Lower bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \geq \exp((\log n)^{1/2-\varepsilon}).$$

Proof. Set $k_0 := \lfloor \exp((\log n)^{1/2-\varepsilon}) \rfloor$.

Step 1: The k_0 -smooth integers in $[1, n]$ are sparse. Let $u_0 = \log n / \log k_0 = (\log n)^{1/2+\varepsilon} (1 + o(1)) \rightarrow \infty$. For all large n we have $u_0 \geq 3$, so Lemma 3 applies (with $x = n$, $y = k_0$) and gives

$$\Psi(n, k_0) \leq n \cdot e^{-\frac{1}{2} u_0 \log u_0}.$$

We compute $\frac{1}{2} u_0 \log u_0$:

$$\frac{1}{2} u_0 \log u_0 = \frac{1}{2} (1 + o(1)) (\log n)^{1/2+\varepsilon} \cdot (1/2 + \varepsilon) \log \log n \geq (\log n)^{1/2+\varepsilon/2}$$

for all large n . Since $\log k_0 = (\log n)^{1/2-\varepsilon}$ and $(\log n)^{1/2+\varepsilon/2} \gg (\log n)^{1/2-\varepsilon}$,

$$e^{-\frac{1}{2} u_0 \log u_0} \leq e^{-(\log n)^{1/2+\varepsilon/2}} \ll \frac{1}{k_0}.$$

Hence $\Psi(n, k_0) \leq n/k_0$ for all sufficiently large n .

Step 2: Pigeonhole produces an empty block. Partition $\{1, \dots, Bk_0\}$ (where $B = \lfloor n/k_0 \rfloor$) into B consecutive blocks $I_i = [(i-1)k_0 + 1, ik_0]$ of length k_0 , for $i = 1, \dots, B$. The number of k_0 -smooth integers in $\{1, \dots, Bk_0\}$ is at most $\Psi(n, k_0)$. From Step 1, $\Psi(n, k_0) \leq n \cdot e^{-(\log n)^{1/2+\varepsilon/2}} = o(n/k_0) = o(B)$. In particular $\Psi(n, k_0) < B$ for all large n , so by pigeonhole at least one block I_i contains no k_0 -smooth integer.

Step 3: The empty block is a k_0 -run. For that I_i : every $j \in I_i$ satisfies $P^+(j) > k_0$. Setting $m = (i-1)k_0 \leq n$, the integers $m+1, \dots, m+k_0$ all have largest prime factor $> k_0$, giving a k_0 -run below n . Hence $k(n) \geq k_0 = e^{(\log n)^{1/2-\varepsilon}(1+o(1))}$. $\square$

Remark 8. A naive Chinese Remainder Theorem construction—choosing distinct primes $q_1, \dots, q_k > k$ and solving $q_j \mid m+j$ by CRT—yields modulus $Q = \prod q_j$ with $\log Q \approx k \log k$. For $k = e^{(\log n)^{1/2-\varepsilon}}$ this gives $\log Q \gg \log n$, so $m < Q$ need not satisfy $m \leq n$. The CRT approach yields only $k(n) \geq (1 - o(1)) \log n / \log \log n$, far weaker.

4 The Upper Bound

Theorem 9 (Upper bound). *For every fixed $\varepsilon > 0$ and all sufficiently large n ,*

$$k(n) \leq \exp((\log n)^{1/2+\varepsilon}).$$

Proof. Set $k_1 := \lfloor \exp((\log n)^{1/2+\varepsilon}) \rfloor$. We show every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer, so no k_1 -run below n exists. We split into two cases.

Case 1: $m \geq k_1$.

Step 1a: Verifying hypothesis (1). We apply Theorem 5 with $x = m$, $y = k_1$, and parameter ε . For $m \geq k_1$ we have $\log m \geq \log k_1 = (\log n)^{1/2+\varepsilon}$, so $\log \log m \leq \log \log n$. We need $\log k_1 \geq (\log \log m)^{5/3+\varepsilon}$, i.e. $(\log n)^{1/2+\varepsilon} \geq (\log \log n)^{5/3+\varepsilon}$. This holds for all large n since any positive power of $\log n$ eventually dominates any power of $\log \log n$. Thus hypothesis (1) holds for all $m \in [k_1, n]$, and Theorem 5 applies.

Step 1b: Smooth integers are dense in $[m+1, m+k_1]$. Set $u_1 = \log m / \log k_1$. Since $m \in [k_1, n]$,

$$u_1 \in \left[1, \frac{\log n}{\log k_1}\right] = [1, (\log n)^{1/2-\varepsilon}].$$

Apply Theorem 5 with $\delta = \frac{1}{2}$. The theorem guarantees the existence of a finite threshold $Y_0 = Y_0(\varepsilon, \frac{1}{2})$, depending only on ε , such that $|\eta(x, y)| \leq \frac{1}{2}$ for all $y \geq Y_0$ and all x satisfying (1). Since $k_1 = \exp((\log n)^{1/2+\varepsilon}) \rightarrow \infty$, there exists $n_0 = n_0(\varepsilon)$ such that $k_1 \geq Y_0$ for all $n \geq n_0$.

For all $n \geq n_0$ and every $m \in [k_1, n]$: hypothesis (1) holds (Step 1a) and $y = k_1 \geq Y_0$, so Theorem 5 gives

$$\Psi(m + k_1, k_1) - \Psi(m, k_1) = k_1 \rho(u_1) (1 + \eta(m, k_1)), \quad |\eta(m, k_1)| \leq \frac{1}{2}.$$

Crucially, Y_0 is independent of $x = m$: the single condition $k_1 \geq Y_0$ suffices for all $m \in [k_1, n]$ simultaneously. This is precisely the independence-of- x stated in Theorem 5. The threshold n_0 therefore depends only on ε , not on m . Hence:

$$\Psi(m + k_1, k_1) - \Psi(m, k_1) \geq \frac{1}{2} k_1 \rho(u_1) \quad \text{for all } m \in [k_1, n]. \quad (2)$$

It remains to show $k_1 \rho(u_1) \geq 2$ uniformly for all $m \in [k_1, n]$. (Note: for $u_1 \in [1, 2]$, $\rho(u_1)$ is bounded below by a positive constant— ρ is continuous, positive, and decreasing with $\rho(1) = 1$ —so $k_1 \rho(u_1) \geq k_1 \rho(2) > 0$ in this range; the lower bound ≥ 2 follows trivially from $k_1 \rightarrow \infty$. For $u_1 > 2$ we argue as follows.) Since ρ is decreasing and $u_1 \leq (\log n)^{1/2-\varepsilon}$,

$$k_1 \rho(u_1) \geq k_1 \rho((\log n)^{1/2-\varepsilon}).$$

Using $\rho(u) = e^{-u \log u (1+o(1))}$ ([4, Cor. III.5.4]):

$$\log(k_1 \rho((\log n)^{1/2-\varepsilon})) = (\log n)^{1/2+\varepsilon} - (1/2-\varepsilon)(\log n)^{1/2-\varepsilon} \log \log n \cdot (1+o(1)).$$

Since $(1/2-\varepsilon)(\log n)^{1/2-\varepsilon} \log \log n = o((\log n)^{1/2+\varepsilon})$, the right side is $(\log n)^{1/2+\varepsilon}(1 - o(1)) \rightarrow +\infty$. Hence there exists $n_1 = n_1(\varepsilon)$ such that for all $n \geq n_1$, $k_1 \rho((\log n)^{1/2-\varepsilon}) \geq 2$. In particular, $k_1 \rho(u_1) \geq 2$ for all $m \in [k_1, n]$ and all $n \geq n_1$.

Substituting into (2):

$$\Psi(m + k_1, k_1) - \Psi(m, k_1) \geq \frac{1}{2} \cdot 2 = 1 \quad \text{for all } m \in [k_1, n] \text{ and all large } n.$$

Case 2: $m < k_1$. Since $m < k_1$, the integer k_1 lies in the interval $[m+1, m+k_1]$ (as $m+1 \leq k_1 \leq m+k_1$). Since all prime factors of k_1 are $\leq k_1$, it is k_1 -smooth.

Conclusion. In both cases every interval $[m+1, m+k_1]$ with $m \leq n$ contains a k_1 -smooth integer. No k_1 -run below n exists, so $k(n) < k_1 \leq e^{(\log n)^{1/2+\varepsilon}}$. $\square$

5 Main Result and Sharpness

Theorem 10 (Main Theorem). *As $n \rightarrow \infty$,*

$$\log k(n) = (\log n)^{1/2+o(1)}.$$

In particular $\log k(n)/(\log n)^{1/2}$ is bounded both above and below by positive constants times arbitrarily small positive powers of $\log n$; this determines the correct order of magnitude on the logarithmic scale.

Proof. Theorems 7 and 9. $\square$

Remark 11 (Heuristic for the leading constant). A heuristic argument suggests that $\log k(n) \sim \left(\frac{1}{2} \log n \cdot \log \log n\right)^{1/2}$. The idea is that the transition in $k(n)$ occurs roughly where the global count $\Psi(n, k)$ crosses n/k ; using $\Psi(n, k) \approx n\rho(u)$ with $u = \log n / \log k$ and $\rho(u) = e^{-u \log u(1+o(1))}$, one solves $e^{-u \log u} \approx e^{-\log k}$ to get $u \log u \approx \log k$. Setting $L = \log k$ and $\ell = \log n$ this gives $L^2 \sim \ell \cdot \frac{1}{2} \log \ell$, hence $c = 1/\sqrt{2}$. However, this heuristic conflates a global sparsity threshold (used in the lower bound) with a local density condition (used in the upper bound), and we do not attempt to make it rigorous here.

Remark 12 (Logical structure of the two bounds). The lower and upper bounds use fundamentally different tools and should not be conflated.

- The *lower bound* (Theorem 7) uses a global sparsity estimate: Lemma 3 gives $\Psi(n, k_0) < n/k_0$, and pigeonhole then forces an empty block of length k_0 . No information about smooth numbers in short intervals is needed.
- The *upper bound* (Theorem 9) uses a local density estimate: Theorem 5 gives a lower bound on the number of k_1 -smooth integers in every short interval $[m+1, m+k_1]$. This requires the full strength of the saddle-point asymptotic and its uniform error bound; the global count $\Psi(n, k_1)$ alone is insufficient.

The two criteria are not equivalent, and the constant $1/\sqrt{2}$ suggested by Remark 11 is not established by this proof.

Remark 13 (Primitive sets). Each k -run $\{m+1, \dots, m+k\}$ is a primitive set (no element divides another): if $a \mid b$ with $a < b$ both in the run, then $b \geq 2a > m+k$. The condition $P^+(a) > k$ gives each element a large prime certificate.

References

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